

YEAR 7 MATHEMATICS

Chapter 8

Area and volume

WORKSHEETS



8

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

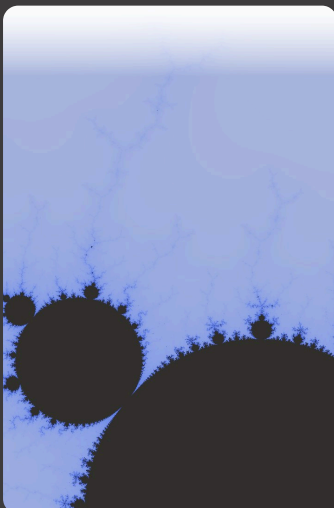
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
8.01	Units of measurement	3		☐
8.02	Perimeter	4		☐
8.03	Circumference of a circle	5		☐
8.04	Area units	6		☐
8.05	Area of squares and rectangles	7		☐
8.06	Area of a triangle	8		☐
8.07	Area of a parallelogram	9		☐
8.08	Area of composite shapes	10		☐
8.09	Prisms, cross-sections and views	11		☐
8.10	Volume units	12		☐
8.11	Volume of a rectangular prism	13		☐
8.12	Volume of a prism	14		☐
8.13	Capacity	15		☐

In this chapter you will

- convert units of length, mass, capacity and time
- find perimeter and circumference
- convert area units and find the area of rectangles, triangles, parallelograms and composite shapes
- draw cross-sections and views of prisms
- convert volume units and find the volume of prisms
- find capacity in litres



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the thin branches (dendrites) that grow from the very top of the set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–29

Convert. Big unit to small unit: multiply. Small to big: divide.

WE DO With your teacher: copy the working.

E1 $5.7 \text{ t} = \underline{\hspace{2cm}} \text{ kg}$

E2 $27\,000 \text{ kg} = \underline{\hspace{2cm}} \text{ t}$

YOU DO Your turn.

1 $57 \text{ m} = \underline{\hspace{2cm}} \text{ mm}$ **57 000 mm**

2 $5\,500 \text{ mm} = \underline{\hspace{2cm}} \text{ m}$

3 $4.8 \text{ km} = \underline{\hspace{2cm}} \text{ m}$

4 $27 \text{ cm} = \underline{\hspace{2cm}} \text{ mm}$

5 $56 \text{ cm} = \underline{\hspace{2cm}} \text{ mm}$

6 $46 \text{ kL} = \underline{\hspace{2cm}} \text{ L}$

7 $19 \text{ m} = \underline{\hspace{2cm}} \text{ mm}$

8 $24\,000 \text{ mL} = \underline{\hspace{2cm}} \text{ L}$

9 $4\,800 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$

10 $48 \text{ mm} = \underline{\hspace{2cm}} \text{ cm}$

11 $22 \text{ mm} = \underline{\hspace{2cm}} \text{ cm}$

12 $3.2 \text{ m} = \underline{\hspace{2cm}} \text{ mm}$

13 $8\,000 \text{ L} = \underline{\hspace{2cm}} \text{ kL}$

14 $42\,000 \text{ kg} = \underline{\hspace{2cm}} \text{ t}$

15 $44\,000 \text{ L} = \underline{\hspace{2cm}} \text{ kL}$

16 $9\,000 \text{ kg} = \underline{\hspace{2cm}} \text{ t}$

17 $1.5 \text{ m} = \underline{\hspace{2cm}} \text{ mm}$

18 $4.9 \text{ m} = \underline{\hspace{2cm}} \text{ cm}$

19 $36\,000 \text{ L} = \underline{\hspace{2cm}} \text{ kL}$

20 $7\,000 \text{ g} = \underline{\hspace{2cm}} \text{ kg}$

21 $57 \text{ L} = \underline{\hspace{2cm}} \text{ mL}$

22 $56\,000 \text{ kg} = \underline{\hspace{2cm}} \text{ t}$

23 $1\,400 \text{ mL} = \underline{\hspace{2cm}} \text{ L}$

24 $4.4 \text{ cm} = \underline{\hspace{2cm}} \text{ mm}$

25 $1 \text{ m} = \underline{\hspace{2cm}} \text{ cm}$

26 $44\,000 \text{ mm} = \underline{\hspace{2cm}} \text{ m}$

27 $21 \text{ cm} = \underline{\hspace{2cm}} \text{ mm}$

28 $35 \text{ km} = \underline{\hspace{2cm}} \text{ m}$

29 $3\,800 \text{ g} = \underline{\hspace{2cm}} \text{ kg}$

Medium

Questions 30–39

Write the rule ($\times$ or $\div$ by what?), then convert.

WE DO With your teacher: copy the working.

E1 $32 \text{ kL} = \underline{\hspace{2cm}} \text{ L}$

E2 $2 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$

YOU DO Your turn.

30 $10\,000 \text{ m} = \underline{\hspace{2cm}} \text{ km}$

m to km: $\div 1\,000$

$10\,000 \div 1\,000 = 10 \text{ km}$

31 $23\,000 \text{ mL} = \underline{\hspace{2cm}} \text{ L}$

mL to L: $\div 1\,000$

$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ L}$

32 $4\,000 \text{ m} = \underline{\hspace{2cm}} \text{ km}$

m to km: $\div$

$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ km}$

33 $6 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$

34 $38\,000 \text{ L} = \underline{\hspace{2cm}} \text{ kL}$

35 $20\,000 \text{ mL} = \underline{\hspace{2cm}} \text{ L}$

36 $2.6 \text{ L} = \underline{\hspace{2cm}} \text{ mL}$

37 $4\,500 \text{ mm} = \underline{\hspace{2cm}} \text{ m}$

38 $460 \text{ mm} = \underline{\hspace{2cm}} \text{ cm}$

39 $4 \text{ t} = \underline{\hspace{2cm}} \text{ kg}$

Challenging

Questions 40–47

Change to the same unit, then answer the question.

WE DO With your teacher: copy the working.

E1 Add 7 m and 910 cm . Give the answer in metres.

E2 A 8 L jug has 461 mL poured out. How much is left, in mL?

YOU DO Your turn.

40 Add 5 m and 518 cm . Give the answer in metres.

$518 \text{ cm} = 5.18 \text{ m}$

$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ m}$

41 A 5 L jug has 535 mL poured out. How much is left, in mL?

$\underline{\hspace{2cm}} \text{ L} = \underline{\hspace{2cm}} \text{ mL}$

$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ mL}$

42 Add 2 m and 351 cm . Give the answer in metres.

43 A 8 L jug has 496 mL poured out. How much is left, in mL?

44 Add 3 m and 925 cm . Give the answer in metres.

45 A 4 L jug has 905 mL poured out. How much is left, in mL?

46 Add 5 m and 188 cm . Give the answer in metres.

47 A 7 L jug has 399 mL poured out. How much is left, in mL?

Easy

Questions 1–20

Find the perimeter: add all the sides.

**WE DO** With your teacher: copy the working.

E1 21 m by 9 m rectangle _____

E2 21 m by 8 m rectangle _____

YOU DO Your turn.1 6 m by 4 m rectangle 20 m

2 19 m by 8 m rectangle _____

3 11 m by 2 m rectangle _____

4 10 m by 9 m rectangle _____

5 21 m by 15 m rectangle _____

6 9 m by 9 m rectangle _____

7 21 m by 7 m rectangle _____

8 18 m by 8 m rectangle _____

9 16 m by 12 m rectangle _____

10 21 m by 10 m rectangle _____

11 9 m by 3 m rectangle _____

12 14 m by 13 m rectangle _____

13 20 m by 2 m rectangle _____

14 18 m by 3 m rectangle _____

15 20 m by 13 m rectangle _____

16 13 m by 8 m rectangle _____

17 24 m by 2 m rectangle _____

18 12 m by 6 m rectangle _____

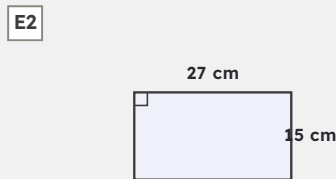
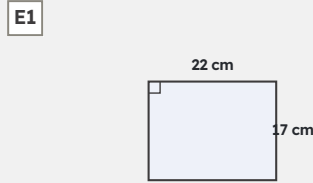
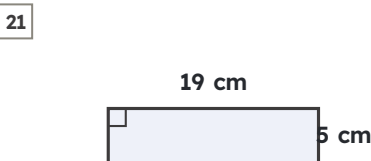
19 23 m by 17 m rectangle _____

20 21 m by 17 m rectangle _____

Medium

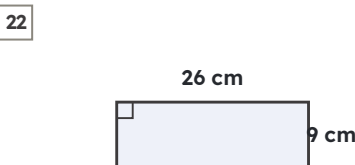
Questions 21–23

Find the perimeter. Write the sum of the sides.

WE DO With your teacher: copy the working.**YOU DO** Your turn.

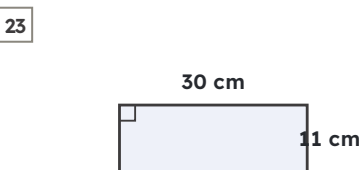
$$19 + 5 + 19 + 5$$

$$= 48 \text{ cm}$$



$$26 + 9 + 26 + 9$$

$$= \text{ } \text{ cm}$$



$$\text{ } + \text{ } + \text{ } + \text{ }$$

$$= \text{ } \text{ cm}$$

Challenging

Questions 24–29

Work backwards from the perimeter.

**WE DO** With your teacher: copy the working.

E1 A rectangle has perimeter 66 m and length 25 m. Find its width.

E2 A square has perimeter 32 cm. Find the side length.

YOU DO Your turn.

24 A rectangle has perimeter 82 m and length 23 m. Find its width.

$$2 \times 23 = 46$$

$$82 - 46 = 36$$

$$\text{width} = 36 \div 2 = 18 \text{ m}$$

25 A square has perimeter 40 cm. Find the side length.

equal sides

$$40 \div 4 = 10 \text{ cm}$$

26 A rectangle has perimeter 44 m and length 16 m. Find its width.

27 A square has perimeter 80 cm. Find the side length.

28 A rectangle has perimeter 26 m and length 8 m. Find its width.

29 A square has perimeter 76 cm. Find the side length.

Easy

Questions 1–29

Find the circumference, $C = \pi d$. Use the π key. Round to 1 decimal place.

WE DO With your teacher: copy the working.

E1 $d = 3 \text{ cm}$

E2 $d = 38 \text{ cm}$

YOU DO Your turn.

- | | | |
|----|---------------------|---------|
| 1 | $d = 21 \text{ cm}$ | 66.0 cm |
| 2 | $d = 27 \text{ cm}$ | |
| 3 | $d = 23 \text{ cm}$ | |
| 4 | $d = 34 \text{ cm}$ | |
| 5 | $d = 11 \text{ cm}$ | |
| 6 | $d = 15 \text{ cm}$ | |
| 7 | $d = 19 \text{ cm}$ | |
| 8 | $d = 18 \text{ cm}$ | |
| 9 | $d = 13 \text{ cm}$ | |
| 10 | $d = 9 \text{ cm}$ | |
| 11 | $d = 6 \text{ cm}$ | |
| 12 | $d = 5 \text{ cm}$ | |
| 13 | $d = 39 \text{ cm}$ | |
| 14 | $d = 20 \text{ cm}$ | |
| 15 | $d = 29 \text{ cm}$ | |
| 16 | $d = 40 \text{ cm}$ | |
| 17 | $d = 32 \text{ cm}$ | |
| 18 | $d = 37 \text{ cm}$ | |
| 19 | $d = 8 \text{ cm}$ | |
| 20 | $d = 12 \text{ cm}$ | |
| 21 | $d = 16 \text{ cm}$ | |
| 22 | $d = 24 \text{ cm}$ | |
| 23 | $d = 33 \text{ cm}$ | |
| 24 | $d = 7 \text{ cm}$ | |
| 25 | $d = 4 \text{ cm}$ | |
| 26 | $d = 2 \text{ cm}$ | |
| 27 | $d = 22 \text{ cm}$ | |
| 28 | $d = 10 \text{ cm}$ | |
| 29 | $d = 36 \text{ cm}$ | |

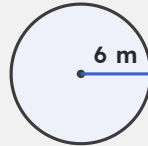
Medium

Questions 30–32

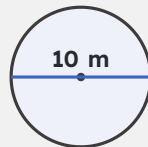
Find the circumference. Write the formula, substitute, then round to 1 d.p.

WE DO With your teacher: copy the working.

E1

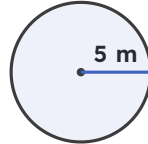


E2



YOU DO Your turn.

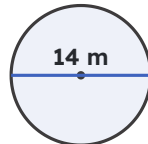
30



$$C = 2\pi r = 2 \times \pi \times 5$$

$$= 31.4 \text{ m}$$

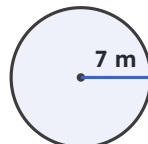
31



$$C = \pi d = \pi \times 14$$

$$= \text{ } \text{ m}$$

32



$$C = \text{ } \pi r = \text{ } \times \pi \times \text{ }$$

$$= \text{ } \text{ m}$$

Challenging

Questions 33–37

Work backwards or find a part of a circle. Show the formula.

WE DO With your teacher: copy the working.

E1 A circle has circumference 53 cm. Find its diameter (1 d.p.).

E2 Find the perimeter of a semicircle with diameter 19 cm (1 d.p.).

YOU DO Your turn.

33 A circle has circumference 72 cm. Find its diameter (1 d.p.).

$$C = \pi d, \text{ so } d = C \div \pi$$

$$d = 72 \div \pi$$

$$= \text{ } \text{ cm}$$

34 Find the perimeter of a semicircle with diameter 17 cm (1 d.p.).

$$\text{curve: } \pi \times \text{ } \div \text{ } = \text{ }$$

$$\text{add the diameter: } \text{ } + \text{ } = \text{ } \text{ cm}$$

35 A circle has circumference 57 cm. Find its diameter (1 d.p.).

36 Find the perimeter of a semicircle with diameter 6 cm (1 d.p.).

37 A circle has circumference 69 cm. Find its diameter (1 d.p.).

Easy

Questions 1–28

Convert. $1 \text{ cm}^2 = 100 \text{ mm}^2$. $1 \text{ m}^2 = 10\,000 \text{ cm}^2$. $1 \text{ ha} = 10\,000 \text{ m}^2$.

WE DO With your teacher: copy the working.

E1 $19 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

E2 $16 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$

YOU DO Your turn.

1 $70\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$ **7 m²**

2 $17 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

3 $2\,500 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

4 $70\,000 \text{ m}^2 = \underline{\hspace{2cm}} \text{ ha}$

5 $5 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$

6 $190\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

7 $1\,900 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

8 $1\,200 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

9 $80\,000 \text{ m}^2 = \underline{\hspace{2cm}} \text{ ha}$

10 $14 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

11 $27 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

12 $17 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

13 $8 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

14 $2\,300 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

15 $18 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

16 $120\,000 \text{ m}^2 = \underline{\hspace{2cm}} \text{ ha}$

17 $3\,000 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

18 $30\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

19 $5 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

20 $180\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

21 $21 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

22 $7 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

23 $220\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

24 $16 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

25 $140\,000 \text{ m}^2 = \underline{\hspace{2cm}} \text{ ha}$

26 $2 \text{ ha} = \underline{\hspace{2cm}} \text{ m}^2$

27 $2\,700 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

28 $20\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

Medium

Questions 29–37

Square the length conversion, then convert the area.

WE DO With your teacher: copy the working.

E1 $2.5 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$

E2 $2 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

YOU DO Your turn.

29 $7 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$

$1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2$

$7 \times 100 = 700 \text{ mm}^2$

30 $5.5 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$

$1 \text{ km}^2 = 1000 \times 1000 = 1\,000\,000 \text{ m}^2$

$\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ m}^2$

31 $2.5 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

$\underline{\hspace{1cm}} \text{ m}^2 = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ cm}^2$

$\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ cm}^2$

32 $8 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$

33 $4 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$

34 $6 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

35 $3 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$

36 $4.5 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$

37 $9 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

Challenging

Questions 38–44

Find the area, then convert it to the unit asked for.

WE DO With your teacher: copy the working.

E1 A paddock is 200 m by 100 m. Find its area in hectares.

E2 A paddock is 800 m by 100 m. Find its area in hectares.

YOU DO Your turn.

38 A paddock is 400 m by 400 m. Find its area in hectares.

$A = 400 \times 400 = 160\,000 \text{ m}^2$

$\div \underline{\hspace{1cm}} : \underline{\hspace{1cm}} \text{ ha}$

39 A paddock is 800 m by 500 m. Find its area in hectares.

$A = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ m}^2$

$\div \underline{\hspace{1cm}} : \underline{\hspace{1cm}} \text{ ha}$

40 A paddock is 300 m by 400 m. Find its area in hectares.

41 A paddock is 200 m by 300 m. Find its area in hectares.

42 A paddock is 900 m by 500 m. Find its area in hectares.

43 A paddock is 200 m by 500 m. Find its area in hectares.

44 A paddock is 700 m by 500 m. Find its area in hectares.

Easy

Questions 1–30

Find the area of the rectangle: $A = \text{length} \times \text{width}$.



WE DO With your teacher: copy the working.

- E1 3 m by 3 m _____
- E2 7 m by 7 m _____

YOU DO Your turn.

- 1 15 m by 5 m 75 m²
- 2 6 m by 3 m _____
- 3 10 m by 3 m _____
- 4 15 m by 10 m _____
- 5 6 m by 6 m _____
- 6 5 m by 3 m _____
- 7 15 m by 13 m _____
- 8 10 m by 10 m _____
- 9 9 m by 5 m _____
- 10 4 m by 4 m _____
- 11 8 m by 6 m _____
- 12 15 m by 15 m _____
- 13 14 m by 10 m _____
- 14 3 m by 2 m _____
- 15 13 m by 10 m _____
- 16 4 m by 2 m _____
- 17 10 m by 7 m _____
- 18 10 m by 8 m _____
- 19 15 m by 12 m _____
- 20 14 m by 13 m _____
- 21 7 m by 6 m _____
- 22 12 m by 4 m _____
- 23 7 m by 2 m _____
- 24 13 m by 9 m _____
- 25 14 m by 11 m _____
- 26 9 m by 7 m _____
- 27 5 m by 5 m _____
- 28 9 m by 6 m _____
- 29 9 m by 3 m _____
- 30 13 m by 3 m _____

Medium

Questions 31–33



Find the area. Write the formula, then substitute.

WE DO With your teacher: copy the working.

E1

E2

YOU DO Your turn.

- 31
-
- $A = lw = 16 \times 8$
 $= 128 \text{ cm}^2$
- 32
-
- $A = lw = 9 \times 3$
 $= \text{ } \text{cm}^2$
- 33
-
- $A = lw = \text{ } \times \text{ } \text{cm}^2$

Challenging

Questions 34–41

Work backwards from the area, or find the area of a square.



WE DO With your teacher: copy the working.

- E1 A rectangle has area 8 m^2 and width 2 m. Find its length.

- E2 A square has sides of 7.5 cm. Find its area.

YOU DO Your turn.

- 34 A rectangle has area 34 m^2 and width 2 m. Find its length.
 $A = lw$, so $34 = l \times 2$
 $l = \text{ } \div \text{ } = \text{ } \text{ m}$
- 35 A square has sides of 14.5 cm. Find its area.
 $A = s^2 = \text{ } \times \text{ } \text{ cm}^2$
- 36 A rectangle has area 78 m^2 and width 6 m. Find its length.

- 37 A square has sides of 9.5 cm. Find its area.

- 38 A rectangle has area 45 m^2 and width 9 m. Find its length.

- 39 A square has sides of 5.5 cm. Find its area.

- 40 A rectangle has area 42 m^2 and width 6 m. Find its length.

- 41 A square has sides of 3.5 cm. Find its area.

Easy

Questions 1–30

Find the area of the triangle:
 $A = \frac{1}{2} \times \text{base} \times \text{height}$.



WE DO With your teacher: copy the working.

E1 $b = 4 \text{ cm}, h = 20 \text{ cm}$ _____

E2 $b = 14 \text{ cm}, h = 2 \text{ cm}$ _____

YOU DO Your turn.

1 $b = 16 \text{ cm}, h = 7 \text{ cm}$ 56 cm²

2 $b = 10 \text{ cm}, h = 3 \text{ cm}$ _____

3 $b = 2 \text{ cm}, h = 18 \text{ cm}$ _____

4 $b = 14 \text{ cm}, h = 18 \text{ cm}$ _____

5 $b = 9 \text{ cm}, h = 20 \text{ cm}$ _____

6 $b = 11 \text{ cm}, h = 16 \text{ cm}$ _____

7 $b = 14 \text{ cm}, h = 8 \text{ cm}$ _____

8 $b = 8 \text{ cm}, h = 14 \text{ cm}$ _____

9 $b = 4 \text{ cm}, h = 4 \text{ cm}$ _____

10 $b = 6 \text{ cm}, h = 15 \text{ cm}$ _____

11 $b = 6 \text{ cm}, h = 7 \text{ cm}$ _____

12 $b = 16 \text{ cm}, h = 15 \text{ cm}$ _____

13 $b = 11 \text{ cm}, h = 2 \text{ cm}$ _____

14 $b = 2 \text{ cm}, h = 8 \text{ cm}$ _____

15 $b = 2 \text{ cm}, h = 9 \text{ cm}$ _____

16 $b = 17 \text{ cm}, h = 12 \text{ cm}$ _____

17 $b = 13 \text{ cm}, h = 12 \text{ cm}$ _____

18 $b = 10 \text{ cm}, h = 11 \text{ cm}$ _____

19 $b = 8 \text{ cm}, h = 8 \text{ cm}$ _____

20 $b = 12 \text{ cm}, h = 9 \text{ cm}$ _____

21 $b = 4 \text{ cm}, h = 11 \text{ cm}$ _____

22 $b = 18 \text{ cm}, h = 3 \text{ cm}$ _____

23 $b = 18 \text{ cm}, h = 9 \text{ cm}$ _____

24 $b = 12 \text{ cm}, h = 8 \text{ cm}$ _____

25 $b = 5 \text{ cm}, h = 12 \text{ cm}$ _____

26 $b = 4 \text{ cm}, h = 18 \text{ cm}$ _____

27 $b = 10 \text{ cm}, h = 18 \text{ cm}$ _____

28 $b = 13 \text{ cm}, h = 16 \text{ cm}$ _____

29 $b = 4 \text{ cm}, h = 12 \text{ cm}$ _____

30 $b = 7 \text{ cm}, h = 16 \text{ cm}$ _____

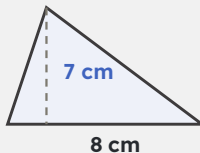
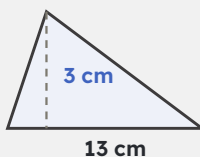
Medium

Questions 31–33

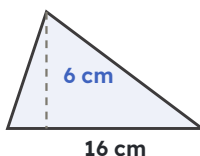
Find the area. The height is at right angles to the base.
 (Not to scale.)



WE DO With your teacher: copy the working.

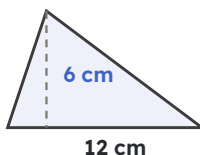
E1**E2**

YOU DO Your turn.

31

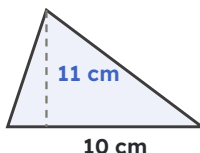
$$A = \frac{1}{2} \times 16 \times 6$$

$$= 48 \text{ cm}^2$$

32

$$A = \frac{1}{2} \times 12 \times 6$$

$$= \text{ } \text{ cm}^2$$

33

$$A = \frac{1}{2} \times \text{ } \times \text{ }$$

$$= \text{ } \text{ cm}^2$$

Challenging

Questions 34–39

Work backwards: find the height or the base.



WE DO With your teacher: copy the working.

E1 A triangle has area 42 cm² and base 6 cm. Find its height.

E2 A triangle has area 104 cm² and base 16 cm. Find its height.

YOU DO Your turn.

34 A triangle has area 90 cm² and base 12 cm. Find its height.

$$90 = \frac{1}{2} \times 12 \times h$$

$$90 = 6 \times h$$

$$h = \text{ } \div \text{ } = \text{ } \text{ cm}$$

35 A triangle has area 28 cm² and base 7 cm. Find its height.

$$\text{ } = \frac{1}{2} \times \text{ } \times h$$

$$\text{ } = \text{ } \times h$$

$$h = \text{ } \div \text{ } = \text{ } \text{ cm}$$

36 A triangle has area 21 cm² and base 7 cm. Find its height.

37 A triangle has area 48 cm² and base 6 cm. Find its height.

38 A triangle has area 60 cm² and base 20 cm. Find its height.

39 A triangle has area 34 cm² and base 17 cm. Find its height.

Easy

Questions 1–29

Find the area of the parallelogram: $A = \text{base} \times \text{height}$.

WE DO With your teacher: copy the working.

E1 $b = 8 \text{ m}, h = 5 \text{ m}$ _____

E2 $b = 9 \text{ m}, h = 7 \text{ m}$ _____

YOU DO Your turn.

1 $b = 6 \text{ m}, h = 6 \text{ m}$ 36 m^2 _____

2 $b = 19 \text{ m}, h = 15 \text{ m}$ _____

3 $b = 3 \text{ m}, h = 7 \text{ m}$ _____

4 $b = 15 \text{ m}, h = 9 \text{ m}$ _____

5 $b = 14 \text{ m}, h = 15 \text{ m}$ _____

6 $b = 19 \text{ m}, h = 2 \text{ m}$ _____

7 $b = 9 \text{ m}, h = 10 \text{ m}$ _____

8 $b = 18 \text{ m}, h = 10 \text{ m}$ _____

9 $b = 18 \text{ m}, h = 8 \text{ m}$ _____

10 $b = 3 \text{ m}, h = 3 \text{ m}$ _____

11 $b = 19 \text{ m}, h = 11 \text{ m}$ _____

12 $b = 11 \text{ m}, h = 5 \text{ m}$ _____

13 $b = 5 \text{ m}, h = 11 \text{ m}$ _____

14 $b = 12 \text{ m}, h = 3 \text{ m}$ _____

15 $b = 16 \text{ m}, h = 12 \text{ m}$ _____

16 $b = 3 \text{ m}, h = 15 \text{ m}$ _____

17 $b = 7 \text{ m}, h = 7 \text{ m}$ _____

18 $b = 6 \text{ m}, h = 15 \text{ m}$ _____

19 $b = 18 \text{ m}, h = 3 \text{ m}$ _____

20 $b = 6 \text{ m}, h = 5 \text{ m}$ _____

21 $b = 12 \text{ m}, h = 5 \text{ m}$ _____

22 $b = 16 \text{ m}, h = 6 \text{ m}$ _____

23 $b = 19 \text{ m}, h = 8 \text{ m}$ _____

24 $b = 11 \text{ m}, h = 14 \text{ m}$ _____

25 $b = 20 \text{ m}, h = 2 \text{ m}$ _____

26 $b = 9 \text{ m}, h = 4 \text{ m}$ _____

27 $b = 13 \text{ m}, h = 4 \text{ m}$ _____

28 $b = 18 \text{ m}, h = 5 \text{ m}$ _____

29 $b = 20 \text{ m}, h = 5 \text{ m}$ _____

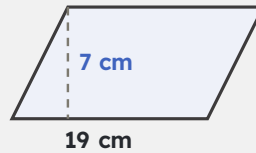
Medium

Questions 30–32

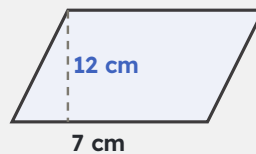
Find the area. Use the perpendicular height, not the slanted side. (Not to scale.)

WE DO With your teacher: copy the working.

E1

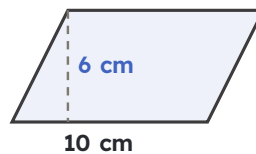


E2



YOU DO Your turn.

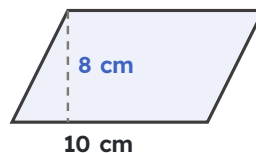
30



$A = bh = 10 \times 6$

$= 60 \text{ cm}^2$

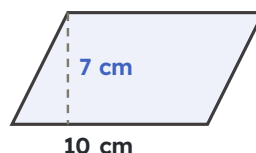
31



$A = bh = 10 \times 8$

$= \text{ } \text{ cm}^2$

32



$A = bh = \text{ } \times \text{ }$

$= \text{ } \text{ cm}^2$

Challenging

Questions 33–37

Work backwards from the area.

WE DO With your teacher: copy the working.

E1 A parallelogram has area 104 m^2 and height 8 m . Find its base.

E2 A parallelogram has area 98 m^2 and height 7 m . Find its base.

YOU DO Your turn.

33 A parallelogram has area 49 m^2 and height 7 m . Find its base.

$49 = b \times 7$

$b = 49 \div 7$

$= \text{ } \text{ m}$

34 A parallelogram has area 120 m^2 and height 8 m . Find its base.

$\text{ } = b \times \text{ }$

$b = \text{ } \div \text{ }$

$= \text{ } \text{ m}$

35 A parallelogram has area 48 m^2 and height 8 m . Find its base.

36 A parallelogram has area 66 m^2 and height 11 m . Find its base.

37 A parallelogram has area 192 m^2 and height 12 m . Find its base.

Easy

Questions 1–29

The shape is two rectangles joined. Add their areas.



WE DO With your teacher: copy the working.

E1 9×4 and 6×7 (m) _____

E2 5×6 and 10×5 (m) _____

YOU DO Your turn.

1 12×4 and 9×3 (m) **75 m²**

2 4×9 and 2×8 (m) _____

3 12×8 and 5×2 (m) _____

4 7×9 and 10×3 (m) _____

5 5×5 and 3×8 (m) _____

6 11×4 and 3×3 (m) _____

7 12×9 and 8×6 (m) _____

8 4×3 and 6×2 (m) _____

9 3×3 and 7×6 (m) _____

10 9×7 and 10×4 (m) _____

11 3×4 and 5×4 (m) _____

12 9×2 and 10×6 (m) _____

13 10×7 and 5×2 (m) _____

14 4×9 and 5×8 (m) _____

15 11×4 and 9×4 (m) _____

16 3×3 and 4×4 (m) _____

17 4×5 and 10×3 (m) _____

18 8×9 and 7×5 (m) _____

19 4×7 and 8×8 (m) _____

20 12×4 and 10×7 (m) _____

21 10×8 and 10×5 (m) _____

22 3×6 and 3×8 (m) _____

23 6×9 and 3×8 (m) _____

24 5×5 and 4×3 (m) _____

25 6×2 and 10×3 (m) _____

26 8×5 and 2×3 (m) _____

27 3×3 and 7×3 (m) _____

28 5×8 and 3×3 (m) _____

29 9×8 and 3×2 (m) _____

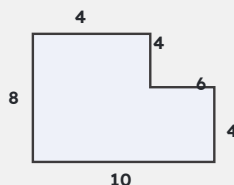
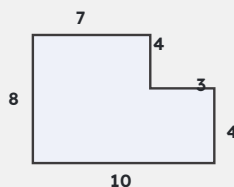
Medium

Questions 30–31

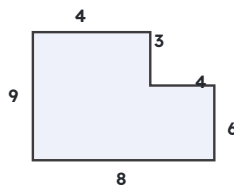
Find the area of the L-shape. Split it into two rectangles. (Not to scale.)



WE DO With your teacher: copy the working.

E1**E2**

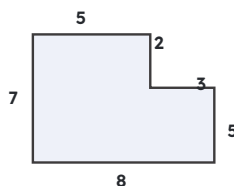
YOU DO Your turn.

30

$4 \times 9 = 36$

$4 \times 6 = 24$

total: 60 cm²

31

$5 \times 7 = 35$

$3 \times 5 = 15$

total: ___ cm²

Challenging

Questions 32–35

Subtract the part cut out, or add a triangle to a rectangle.



WE DO With your teacher: copy the working.

E1 A 15 cm × 12 cm rectangle has a 3 cm × 3 cm hole cut out. Find the area left.

E2 A house wall is a 18 m × 9 m rectangle with a triangle on top (base 18 m, height 3 m). Find its area.

YOU DO Your turn.

32 A 8 cm × 14 cm rectangle has a 3 cm × 3 cm hole cut out. Find the area left.

big: $8 \times 14 = 112$

hole: $3 \times 3 = 9$

___ - ___ = ___ cm²

33 A house wall is a 13 m × 11 m rectangle with a triangle on top (base 13 m, height 4 m). Find its area.

rectangle: $___ \times ___ = ___$

triangle: $\frac{1}{2} \times ___ \times ___ = ___$

total: $___ \text{ m}^2$

34 A 19 cm × 10 cm rectangle has a 4 cm × 4 cm hole cut out. Find the area left.

35 A house wall is a 16 m × 12 m rectangle with a triangle on top (base 16 m, height 7 m). Find its area.

Easy

Questions 1–21

Name the shape of the cross-section of each prism.



WE DO With your teacher: copy the working.

E1 triangular prism

E2 rectangular prism

YOU DO Your turn.

1 pentagonal prism pentagon

2 hexagonal prism

3 octagonal prism

4 cube

5 cylinder

6 a Toblerone box

7 a cereal box

8 a six-sided pencil

9 square prism

10 a tent with triangle ends

11 a can of soup

12 a brick

13 a honeycomb cell

14 a stop sign, as a prism

15 decagonal prism

16 a tissue box

17 heptagonal prism

18 a 50c coin

19 a water pipe

20 a ruler

21 a dice

Medium

Questions 22–27



A prism has n sides on its cross-section. Faces = $n + 2$, vertices = $2n$, edges = $3n$.

WE DO With your teacher: copy the working.

E1 cross-section: triangle

E2 cross-section: square

YOU DO Your turn.

22 cross-section: pentagon

$$n = 5: F = 5 + 2 = 7$$

$$V = 2 \times 5 = 10, E = 3 \times 5 = 15$$

23 cross-section: hexagon

$$n = 6: F = 6 + 2 = 8$$

$$V = ___ \times ___ = ___, E = ___ \times ___ = ___$$

24 cross-section: heptagon

$$n = ___: F = ___ + ___ = ___$$

$$V = ___ \times ___ = ___, E = ___ \times ___ = ___$$

25 cross-section: octagon

26 cross-section: decagon

27 cross-section: 12-sided polygon

Challenging

Questions 28–31

Describe the views of each solid. (Front: what you see from in front; top: from above.)



WE DO With your teacher: copy the working.

E1 a cylinder standing up

E2 a square pyramid

YOU DO Your turn.

28 a triangular prism lying on a rectangular face

front view: rectangle

top view: _____

29 a cone standing on its base

front view: _____

top view: _____

30 a cube

31 a rectangular prism standing on end

Easy

Questions 1–29

Convert. $1 \text{ cm}^3 = 1000 \text{ mm}^3$.
 $1 \text{ cm}^3 = 1 \text{ mL}$. $1000 \text{ cm}^3 = 1 \text{ L}$.

WE DO With your teacher: copy the working.

E1 $140 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

E2 $36\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

YOU DO Your turn.

1 $560 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$ **560 mL**

2 $22\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

3 $190 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

4 $340 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

5 $44\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

6 $42\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

7 $4\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

8 $250 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

9 $31 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

10 $59\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

11 $51\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

12 $290 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

13 $8\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

14 $260 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

15 $8 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

16 $22 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

17 $49 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

18 $30\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

19 $12 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

20 $38\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

21 $20\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

22 $50\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

23 $540 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

24 $54\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

25 $510 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

26 $370 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

27 $33\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

28 $52\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ L}$

29 $550 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mL}$

Medium

Questions 30–39

Cube the length conversion, then convert the volume.

WE DO With your teacher: copy the working.

E1 $4 \text{ m}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

E2 $9 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

YOU DO Your turn.

30 $3 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

$1 \text{ cm}^3 = 10 \times 10 \times 10 = 1\,000 \text{ mm}^3$

$3 \times 1\,000 = 3\,000 \text{ mm}^3$

31 $3 \text{ m}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

$1 \text{ m}^3 = 100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3$

$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ cm}^3$

32 $7 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

$\underline{\hspace{2cm}} \text{ cm}^3 = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ mm}^3$

$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ mm}^3$

33 $6 \text{ m}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

34 $6 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

35 $9 \text{ m}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

36 $5 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

37 $4 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

38 $8 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

39 $7 \text{ m}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

Challenging

Questions 40–47

Convert between volume and capacity. Show each step.

WE DO With your teacher: copy the working.

E1 A tank holds 8.5 m^3 of water. How many litres is that?

E2 A tank holds 1 m^3 of water. How many litres is that?

YOU DO Your turn.

40 A tank holds 8 m^3 of water. How many litres is that?

$1 \text{ m}^3 = 1000 \text{ L}$

$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ L}$

41 A tank holds 3 m^3 of water. How many litres is that?

$\underline{\hspace{2cm}} \text{ m}^3 = \underline{\hspace{2cm}} \text{ L}$

$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ L}$

42 A tank holds 4 m^3 of water. How many litres is that?

43 A tank holds 6.5 m^3 of water. How many litres is that?

44 A tank holds 9.5 m^3 of water. How many litres is that?

45 A tank holds 2.5 m^3 of water. How many litres is that?

46 A tank holds 7.5 m^3 of water. How many litres is that?

47 A tank holds 1.5 m^3 of water. How many litres is that?

8.11 Volume of a rectangular prism

YEAR 7 · CHAPTER 8: AREA AND VOLUME

Easy

Questions 1–30

Find the volume: $V = \text{length} \times \text{width} \times \text{height}$.

WE DO With your teacher: copy the working.

E1 $4 \text{ cm} \times 9 \text{ cm} \times 2 \text{ cm}$ _____

E2 $8 \text{ cm} \times 9 \text{ cm} \times 7 \text{ cm}$ _____

YOU DO Your turn.

1 $6 \text{ cm} \times 3 \text{ cm} \times 4 \text{ cm}$ 72 cm^3

2 $3 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$ _____

3 $6 \text{ cm} \times 5 \text{ cm} \times 4 \text{ cm}$ _____

4 $3 \text{ cm} \times 8 \text{ cm} \times 3 \text{ cm}$ _____

5 $5 \text{ cm} \times 3 \text{ cm} \times 8 \text{ cm}$ _____

6 $12 \text{ cm} \times 8 \text{ cm} \times 6 \text{ cm}$ _____

7 $9 \text{ cm} \times 9 \text{ cm} \times 9 \text{ cm}$ _____

8 $3 \text{ cm} \times 8 \text{ cm} \times 6 \text{ cm}$ _____

9 $12 \text{ cm} \times 6 \text{ cm} \times 4 \text{ cm}$ _____

10 $4 \text{ cm} \times 7 \text{ cm} \times 7 \text{ cm}$ _____

11 $5 \text{ cm} \times 3 \text{ cm} \times 4 \text{ cm}$ _____

12 $2 \text{ cm} \times 8 \text{ cm} \times 9 \text{ cm}$ _____

13 $7 \text{ cm} \times 8 \text{ cm} \times 8 \text{ cm}$ _____

14 $5 \text{ cm} \times 9 \text{ cm} \times 4 \text{ cm}$ _____

15 $4 \text{ cm} \times 5 \text{ cm} \times 7 \text{ cm}$ _____

16 $11 \text{ cm} \times 9 \text{ cm} \times 7 \text{ cm}$ _____

17 $11 \text{ cm} \times 7 \text{ cm} \times 9 \text{ cm}$ _____

18 $6 \text{ cm} \times 8 \text{ cm} \times 4 \text{ cm}$ _____

19 $6 \text{ cm} \times 4 \text{ cm} \times 8 \text{ cm}$ _____

20 $7 \text{ cm} \times 7 \text{ cm} \times 8 \text{ cm}$ _____

21 $4 \text{ cm} \times 10 \text{ cm} \times 2 \text{ cm}$ _____

22 $10 \text{ cm} \times 10 \text{ cm} \times 3 \text{ cm}$ _____

23 $9 \text{ cm} \times 10 \text{ cm} \times 6 \text{ cm}$ _____

24 $3 \text{ cm} \times 9 \text{ cm} \times 9 \text{ cm}$ _____

25 $2 \text{ cm} \times 7 \text{ cm} \times 2 \text{ cm}$ _____

26 $4 \text{ cm} \times 8 \text{ cm} \times 4 \text{ cm}$ _____

27 $8 \text{ cm} \times 10 \text{ cm} \times 3 \text{ cm}$ _____

28 $8 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm}$ _____

29 $7 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$ _____

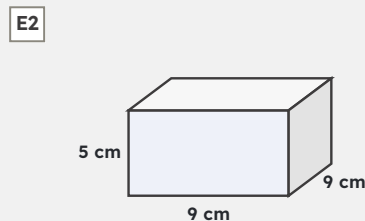
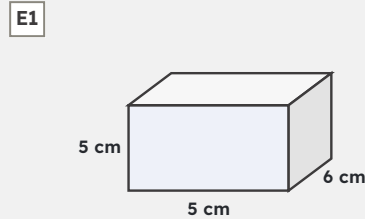
30 $8 \text{ cm} \times 6 \text{ cm} \times 9 \text{ cm}$ _____

Medium

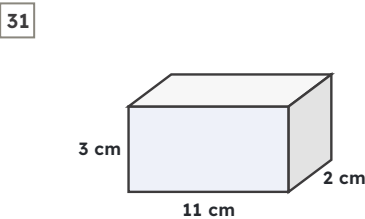
Questions 31–33

Find the volume. Write the formula, then substitute. (Not to scale.)

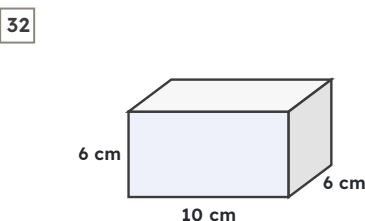
WE DO With your teacher: copy the working.



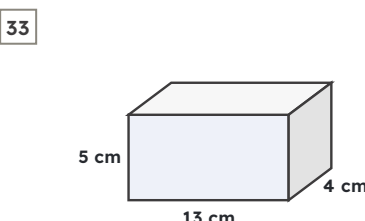
YOU DO Your turn.



$V = lwh = 11 \times 2 \times 3$
 $= 66 \text{ cm}^3$



$V = lwh = 10 \times 6 \times 6$
 $= \underline{\hspace{1cm}} \text{ cm}^3$



$V = lwh = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$
 $= \underline{\hspace{1cm}} \text{ cm}^3$

Challenging

Questions 34–38

Work backwards from the volume.

WE DO With your teacher: copy the working.

E1 A box has volume 192 cm^3 , length 12 cm and width 8 cm . Find its height.

E2 A box has volume 84 cm^3 , length 7 cm and width 2 cm . Find its height.

YOU DO Your turn.

34 A box has volume 448 cm^3 , length 8 cm and width 8 cm . Find its height.

$448 = 8 \times 8 \times h$

$448 = 64 \times h$

$h = \underline{\hspace{1cm}} \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ cm}$

35 A box has volume 198 cm^3 , length 11 cm and width 3 cm . Find its height.

$\underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times h$

$\underline{\hspace{1cm}} = \underline{\hspace{1cm}} \times h$

$h = \underline{\hspace{1cm}} \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ cm}$

36 A box has volume 120 cm^3 , length 8 cm and width 3 cm . Find its height.

37 A box has volume 360 cm^3 , length 10 cm and width 4 cm . Find its height.

38 A box has volume 210 cm^3 , length 7 cm and width 5 cm . Find its height.

Easy

Questions 1–28

Find the volume of the prism: $V = \text{area of cross-section} \times \text{length}$.

WE DO With your teacher: copy the working.

E1 $A = 19 \text{ cm}^2$, $l = 4 \text{ cm}$ _____

E2 $A = 5 \text{ cm}^2$, $l = 8 \text{ cm}$ _____

YOU DO Your turn.

1 $A = 9 \text{ cm}^2$, $l = 3 \text{ cm}$ **27 cm³**

2 $A = 6 \text{ cm}^2$, $l = 10 \text{ cm}$ _____

3 $A = 44 \text{ cm}^2$, $l = 6 \text{ cm}$ _____

4 $A = 34 \text{ cm}^2$, $l = 5 \text{ cm}$ _____

5 $A = 21 \text{ cm}^2$, $l = 4 \text{ cm}$ _____

6 $A = 16 \text{ cm}^2$, $l = 9 \text{ cm}$ _____

7 $A = 55 \text{ cm}^2$, $l = 6 \text{ cm}$ _____

8 $A = 30 \text{ cm}^2$, $l = 3 \text{ cm}$ _____

9 $A = 26 \text{ cm}^2$, $l = 12 \text{ cm}$ _____

10 $A = 38 \text{ cm}^2$, $l = 5 \text{ cm}$ _____

11 $A = 3 \text{ cm}^2$, $l = 4 \text{ cm}$ _____

12 $A = 29 \text{ cm}^2$, $l = 7 \text{ cm}$ _____

13 $A = 52 \text{ cm}^2$, $l = 3 \text{ cm}$ _____

14 $A = 18 \text{ cm}^2$, $l = 11 \text{ cm}$ _____

15 $A = 41 \text{ cm}^2$, $l = 5 \text{ cm}$ _____

16 $A = 25 \text{ cm}^2$, $l = 5 \text{ cm}$ _____

17 $A = 17 \text{ cm}^2$, $l = 6 \text{ cm}$ _____

18 $A = 38 \text{ cm}^2$, $l = 7 \text{ cm}$ _____

19 $A = 16 \text{ cm}^2$, $l = 4 \text{ cm}$ _____

20 $A = 26 \text{ cm}^2$, $l = 9 \text{ cm}$ _____

21 $A = 56 \text{ cm}^2$, $l = 10 \text{ cm}$ _____

22 $A = 21 \text{ cm}^2$, $l = 7 \text{ cm}$ _____

23 $A = 6 \text{ cm}^2$, $l = 4 \text{ cm}$ _____

24 $A = 48 \text{ cm}^2$, $l = 11 \text{ cm}$ _____

25 $A = 55 \text{ cm}^2$, $l = 9 \text{ cm}$ _____

26 $A = 16 \text{ cm}^2$, $l = 2 \text{ cm}$ _____

27 $A = 51 \text{ cm}^2$, $l = 7 \text{ cm}$ _____

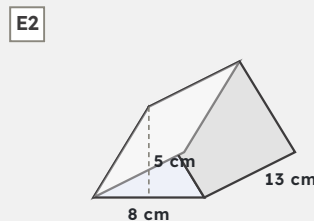
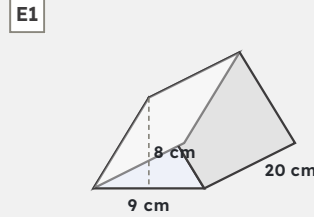
28 $A = 20 \text{ cm}^2$, $l = 7 \text{ cm}$ _____

Medium

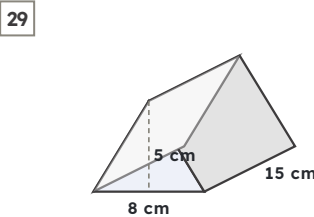
Questions 29–31

Find the area of the triangle, then the volume. (Not to scale.)

WE DO With your teacher: copy the working.

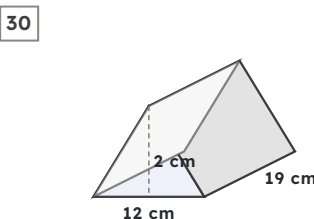


YOU DO Your turn.



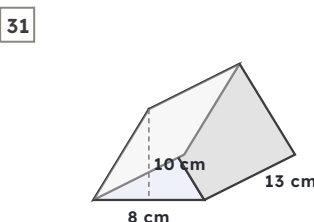
$$A = \frac{1}{2} \times 8 \times 5 = 20$$

$$V = 20 \times 15 = 300 \text{ cm}^3$$



$$A = \frac{1}{2} \times 12 \times 2 = 12$$

$$V = \text{ } \times \text{ } = \text{ } \text{ cm}^3$$



$$A = \frac{1}{2} \times \text{ } \times \text{ } = \text{ }$$

$$V = \text{ } \times \text{ } = \text{ } \text{ cm}^3$$

Challenging

Questions 32–36

Work backwards from the volume.

WE DO With your teacher: copy the working.

E1 A prism has volume 330 m^3 and cross-section area 22 m^2 . Find its length.

E2 A prism has volume 95 m^3 and cross-section area 19 m^2 . Find its length.

YOU DO Your turn.

32 A prism has volume 360 m^3 and cross-section area 40 m^2 . Find its length.

$$360 = 40 \times \text{length}$$

$$\text{length} = 360 \div 40$$

$$= \text{ } \text{ m}$$

33 A prism has volume 90 m^3 and cross-section area 10 m^2 . Find its length.

$$\text{ } = \text{ } \times \text{length}$$

$$\text{length} = \text{ } \div \text{ }$$

$$= \text{ } \text{ m}$$

34 A prism has volume 15 m^3 and cross-section area 5 m^2 . Find its length.

35 A prism has volume 145 m^3 and cross-section area 29 m^2 . Find its length.

36 A prism has volume 165 m^3 and cross-section area 15 m^2 . Find its length.

Easy

Questions 1–30

Convert. 1 L = 1000 mL. 1 kL = 1000 L.



WE DO With your teacher: copy the working.

E1 1 500 mL = ____ L

E2 5 kL = ____ L

YOU DO Your turn.

1 11 L = ____ mL **11 000 mL**

2 250 L = ____ kL

3 2.5 kL = ____ L

4 1 000 L = ____ kL

5 8.5 L = ____ mL

6 18 000 L = ____ kL

7 16 500 mL = ____ L

8 1 L = ____ mL

9 9 000 mL = ____ L

10 14 000 L = ____ kL

11 11.5 kL = ____ L

12 19 000 L = ____ kL

13 0.5 kL = ____ L

14 5 500 mL = ____ L

15 1.5 L = ____ mL

16 35 000 mL = ____ L

17 21 kL = ____ L

18 8 000 L = ____ kL

19 2 kL = ____ L

20 1 750 L = ____ kL

21 1.75 kL = ____ L

22 7 000 mL = ____ L

23 31 kL = ____ L

24 29 000 L = ____ kL

25 750 L = ____ kL

26 31 000 L = ____ kL

27 28 000 mL = ____ L

28 6 000 mL = ____ L

29 8 500 L = ____ kL

30 4.5 kL = ____ L

Medium

Questions 31–40



Find the volume in cm^3 , then the capacity in litres (1000 $\text{cm}^3 = 1 \text{ L}$).

WE DO With your teacher: copy the working.

E1 a box 30 cm × 10 cm × 15 cm

E2 a box 20 cm × 10 cm × 15 cm

YOU DO Your turn.

31 a box 30 cm × 10 cm × 10 cm

$V = 30 \times 10 \times 10 = 3\,000 \text{ cm}^3$

$\div 1000 = 3 \text{ L}$

32 a box 20 cm × 20 cm × 10 cm

$V = 20 \times 20 \times 10 = 4\,000 \text{ cm}^3$

$\div ______ = ______ \text{ L}$

33 a box 50 cm × 20 cm × 10 cm

$V = ______ \times ______ \times ______ = ______ \text{ cm}^3$

$\div ______ = ______ \text{ L}$

34 a box 20 cm × 10 cm × 10 cm

35 a box 20 cm × 30 cm × 25 cm

36 a box 40 cm × 20 cm × 25 cm

37 a box 40 cm × 50 cm × 20 cm

38 a box 60 cm × 50 cm × 20 cm

39 a box 30 cm × 50 cm × 25 cm

40 a box 50 cm × 50 cm × 20 cm

Challenging

Questions 41–46

Solve the capacity problem. Change to the same unit first.



WE DO With your teacher: copy the working.

E1 How many 500 mL cups can be filled from 3 L of juice?

E2 How many 750 mL cups can be filled from 4.5 L of juice?

YOU DO Your turn.

41 How many 250 mL cups can be filled from 9 L of juice?

$9 \text{ L} = 9\,000 \text{ mL}$

$9\,000 \div 250 = 36$

____ cups

42 How many 500 mL cups can be filled from 1.5 L of juice?

____ L = ____ mL

____ $\div$ ____ = ____

____ cups

43 How many 200 mL cups can be filled from 6 L of juice?

44 How many 750 mL cups can be filled from 6 L of juice?

45 How many 750 mL cups can be filled from 1.5 L of juice?

46 How many 300 mL cups can be filled from 1.5 L of juice?